# REDUCEDNESS AND GEOMETRIC REDUCEDNESS ARE PRESERVED BY A PRODUCT WITH A REDUCED OR GEOMETRICALLY REDUCED FACTOR

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed. The statement's last item reads, verbatim, 'Add more here.' — an author placeholder left in the source; it is reproduced as written and not removed, since the delivered material must be the author's text.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

*Editorial note.* Same-chapter references are located at their \label lines inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

*Editorial note.* Overlap pre-check: this label occurs in no other staged chapter (run with agent-authoring/stk/card_plan.py varieties before opening).

*Allowed standard prerequisites (editorial).*

- Geometric reducedness at a point and reducedness after a reduced base change (scheme form). Same chapter, Lemmas lemma-geometrically-reduced-at-point (source line 354, Stacks tag 035W) and lemma-geometrically-reduced (source line 418, Stacks tag 035X); invoked at proof lines 541 and 544. Supply the scheme-level statements to which the algebra lemma is applied in parts (1) and (2).
- Reducedness is preserved by tensor product with a reduced algebra over a field, in the geometrically reduced sense. Algebra, Lemma labelled algebra-lemma-geometrically-reduced-any-reduced-base-change (canonical Stacks tag 034N); invoked at proof lines 542-543 and 545-546. Supplies the algebraic content of parts (1) and (2), i.e. that reducedness survives the base change.
- Over a perfect field reduced implies geometrically reduced. Same chapter, Lemma labelled lemma-perfect-reduced (source line 401, Stacks tag 020I); invoked at proof line 551. Supplies part (4) from part (3) when k is perfect.

*Source reference.* Original lemma 035W: sources/stacks-varieties/pinned-original-source.tex, source lines 354–354. Same-chapter pointer invoked at proof line 541 for the point-wise input of part (1); the combination of the two cited inputs is retained in the counted span.

*Source reference.* Original lemma 034N: sources/stacks-chow/tags.txt, source lines 4098–4098. Cross-chapter standard prerequisite invoked twice, at proof lines 542-543 and 545-546, for the algebraic content of parts (1) and (2).

---

*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 035X: sources/stacks-varieties/pinned-original-source.tex, source lines 418–418. Same-chapter pointer invoked at proof lines 544 and 550 for the global input of parts (2) and (3).

*Source reference.* Original lemma 020I: sources/stacks-varieties/pinned-original-source.tex, source lines 401–401. Same-chapter pointer invoked at proof line 551 to upgrade reducedness to geometric reducedness over a perfect field, giving part (4) from part (3).

**Lemma 0.1.** *Let $k$ be a field. Let $X$, $Y$ be schemes over $k$.*

(1) *If $X$ is geometrically reduced at $x$, and $Y$ reduced, then $X \times_k Y$ is reduced at every point lying over $x$.*
(2) *If $X$ geometrically reduced over $k$ and $Y$ reduced, then $X \times_k Y$ is reduced.*
(3) *If $X$ and $Y$ are geometrically reduced over $k$, then $X \times_k Y$ is geometrically reduced.*
(4) *If $k$ is perfect and $X$ and $Y$ are reduced, then $X \times_k Y$ is reduced.*
(5) *Add more here.*

*Proof.* To prove (1) combine Lemma 035W with Algebra, Lemma 034N. To prove (2) combine Lemma 035X with Algebra, Lemma 034N. To prove (3) note that $(X \times_k Y)_{\overline{k}} = X_{\overline{k}} \times_{\overline{k}} Y_{\overline{k}}$ and use (2) as well as Lemma 035X. To prove (4) use (3) combined with Lemma 020I. □